DATA ANALYTICS PORTFOLIO

Health Insurance Claims Analysis

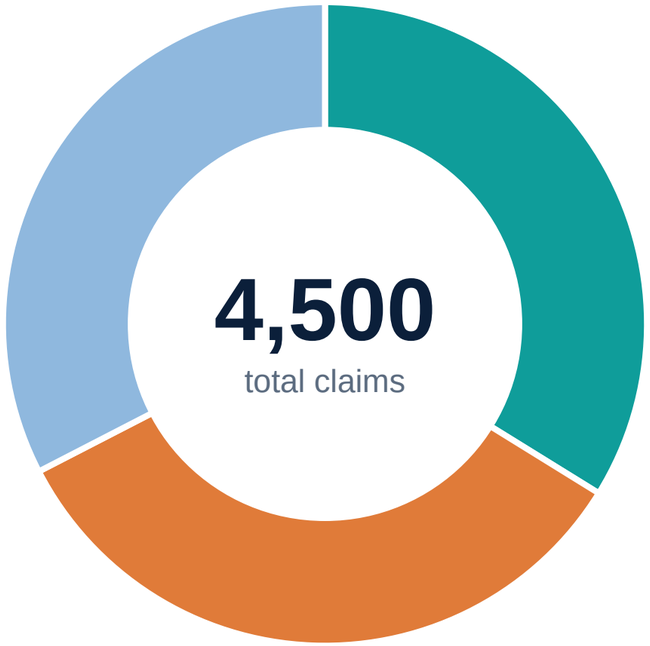
Exploring Claim Rejection Patterns Using SQL

4,500 Claims Analysed | **PostgreSQL** | **Data Analytics Portfolio Project**

SQL Project

CLAIM STATUS OVERVIEW

What Does the Dataset Reveal?



Claim Status	Claims	Percentage
Approved	1,522	33.82%
Denied	1,512	33.60%
Pending	1,466	32.58%

4,500
total claims

1,512
denied claims

33.60%
denial share

Approximately one-third of the claims in the dataset were denied, making claim rejection an important area for further investigation.

WHERE DO REJECTIONS OCCUR?

Exploring Claim Rejection Patterns

Denied Claims by Claim Type

- **Outpatient recorded the highest denied-claim count.**

Denied Claims by Submission Method

- **Online recorded the highest denied-claim count.**

These are denied-claim counts, not denial rates. Total claim volumes within each category must be considered for fair comparisons.

PROVIDER-LEVEL ANALYSIS

Examining Provider-Related Patterns

Provider Specialty – Denied Claims

- **Pediatrics recorded the highest denied-claim count among the listed specialties.**

Provider Location – Top 10 Results

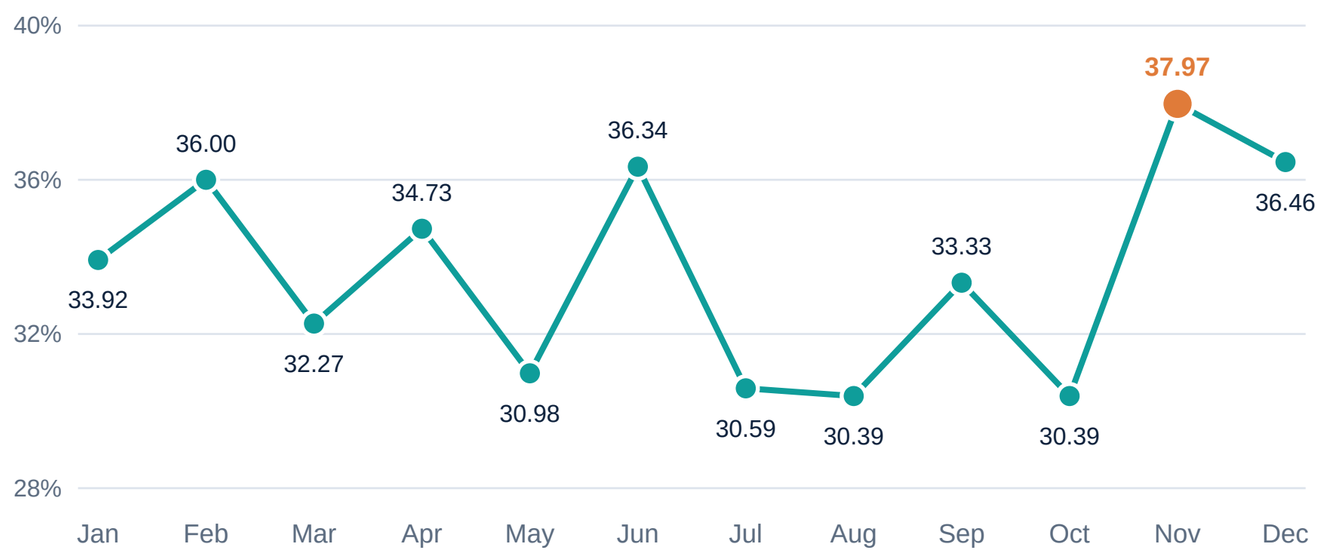
#	Location	Denied claims
1	Port Lisa	4
2	Lake James	3
3	Port Thomas	3
4	East David	3
5	Jessicaburgh	3
6	East Michael	3
7	North Michael	3
8	Ashleyberg	3
9	Christinechester	2
10	Port William	2

Location-level counts are very small and should not be interpreted as evidence of higher rejection risk.

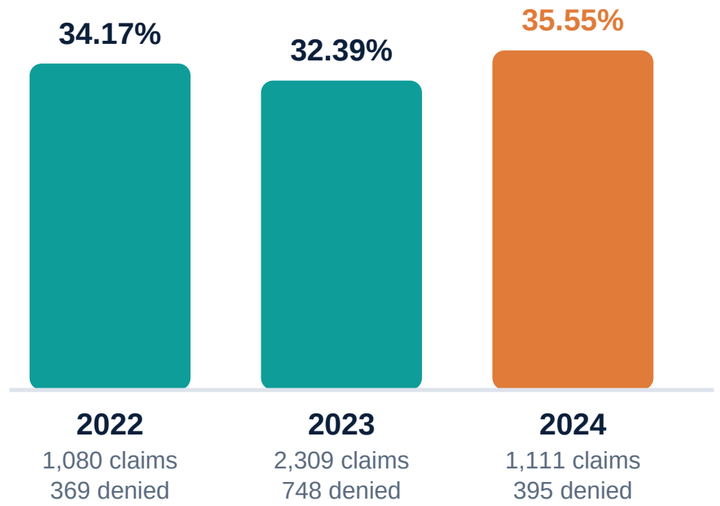
WHEN DO REJECTIONS INCREASE?

Understanding Claim Rejection Trends

Monthly Denial Percentage



Yearly Denial Percentage



November had the highest monthly denial percentage: 37.97%.

2024 had the highest yearly denial percentage: 35.55%.

2023 had the highest total claim volume: 2,309.

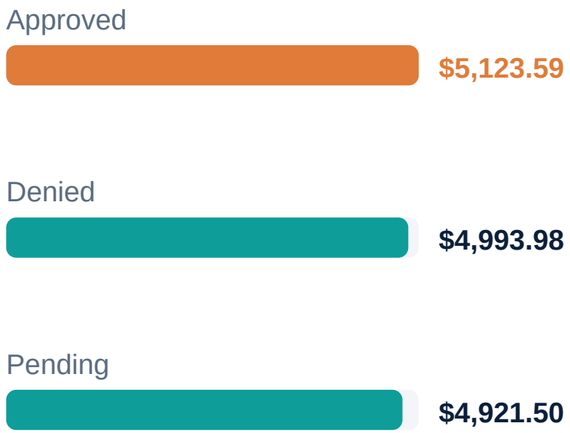
Monthly figures aggregate the same calendar month across all years, rather than representing a chronological monthly trend.

NUMERICAL AND DEMOGRAPHIC INSIGHTS

What Do the Numbers Tell Us?

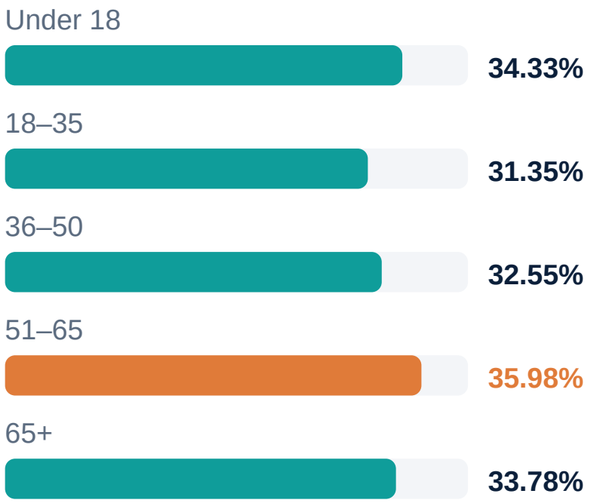
1. Claim Amount by Status

- **Approved claims had the highest average claim amount.**



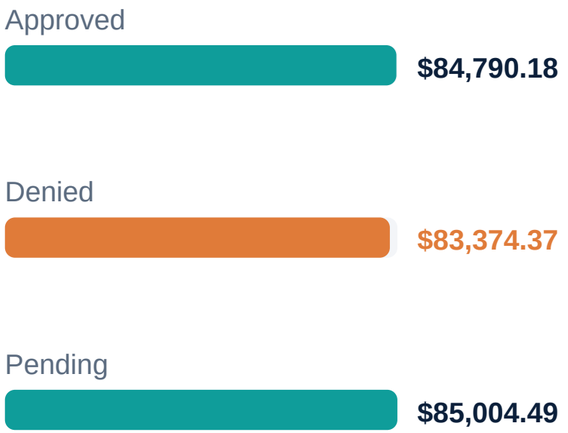
2. Denial Rate by Age Group

- **51–65 had the highest rate; 18–35 the lowest.**



3. Patient Income by Status

- **Denied claims had the lowest average patient income.**



These are observed associations and do not establish the causes of claim rejection. Currency in USD.

CONCLUSION AND WHAT'S NEXT

From SQL Analysis to Business Intelligence

Key Takeaways

1. 33.60% of claims were denied.
2. Rejection counts varied across claim and provider categories.
3. November had the highest monthly denial percentage.
4. The 51–65 age group had the highest denial rate.
5. Further investigation is required to understand the underlying reasons for rejection.

Data Limitations

- No specific denial reasons.
- No actual payout amounts.
- No claim-processing duration information.
- No confirmed fraud labels.

NEXT STEP

POWER BI DASHBOARD – IN PROGRESS

Currently working on transforming these SQL findings into an interactive Power BI dashboard to communicate insights through visual storytelling.

Turning raw data into meaningful business insights.

Tools Used: PostgreSQL | Power BI (In Progress)